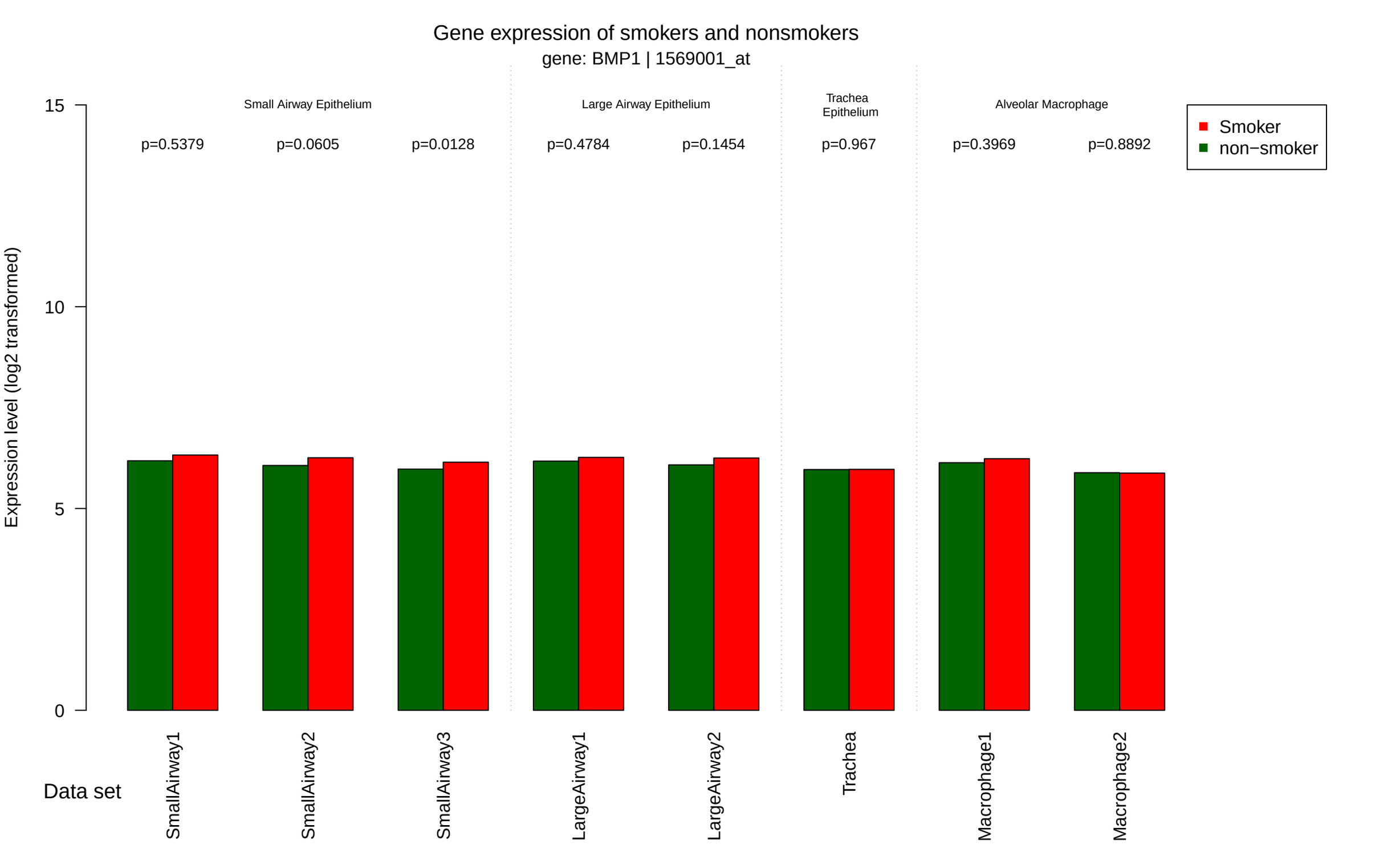
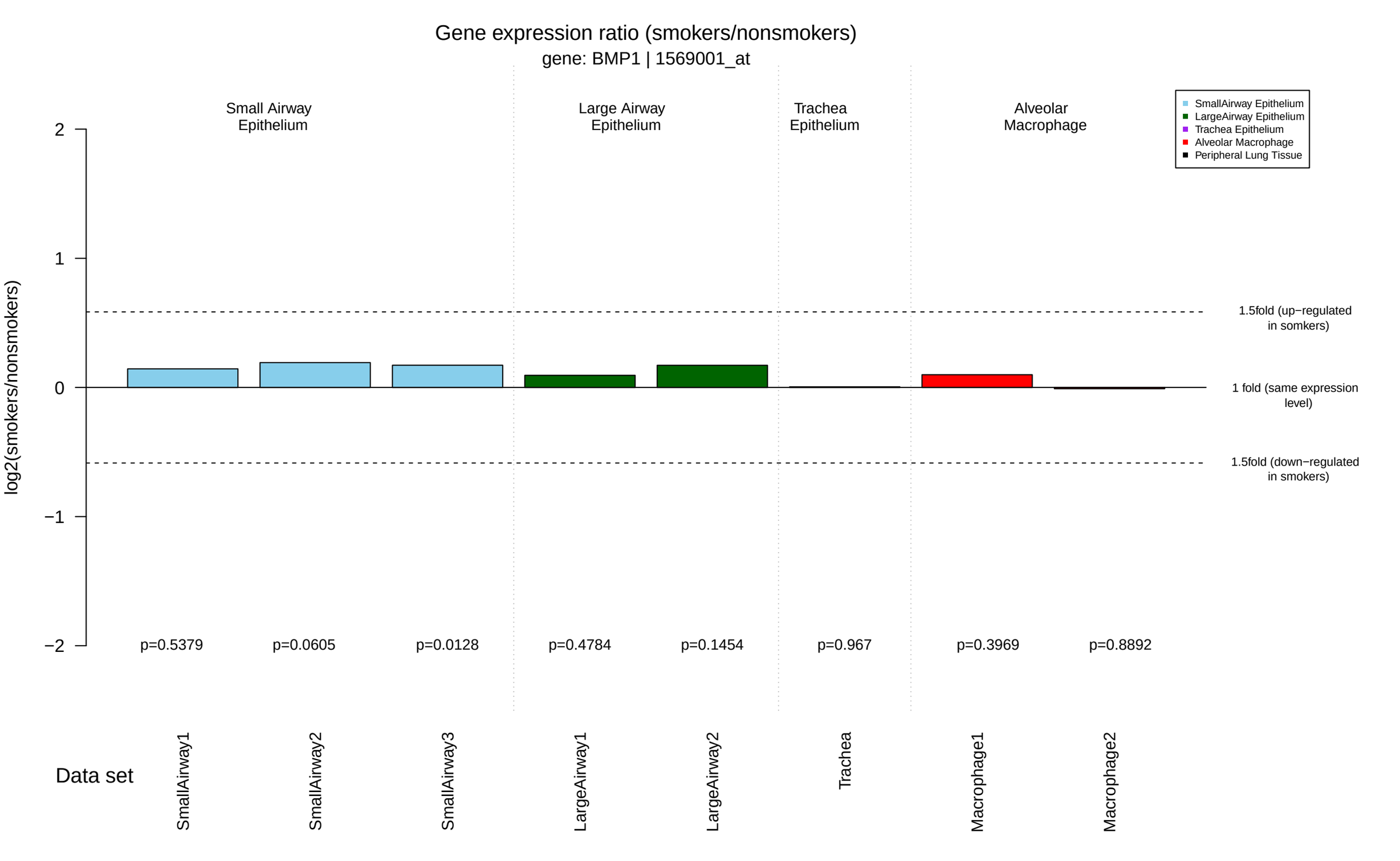
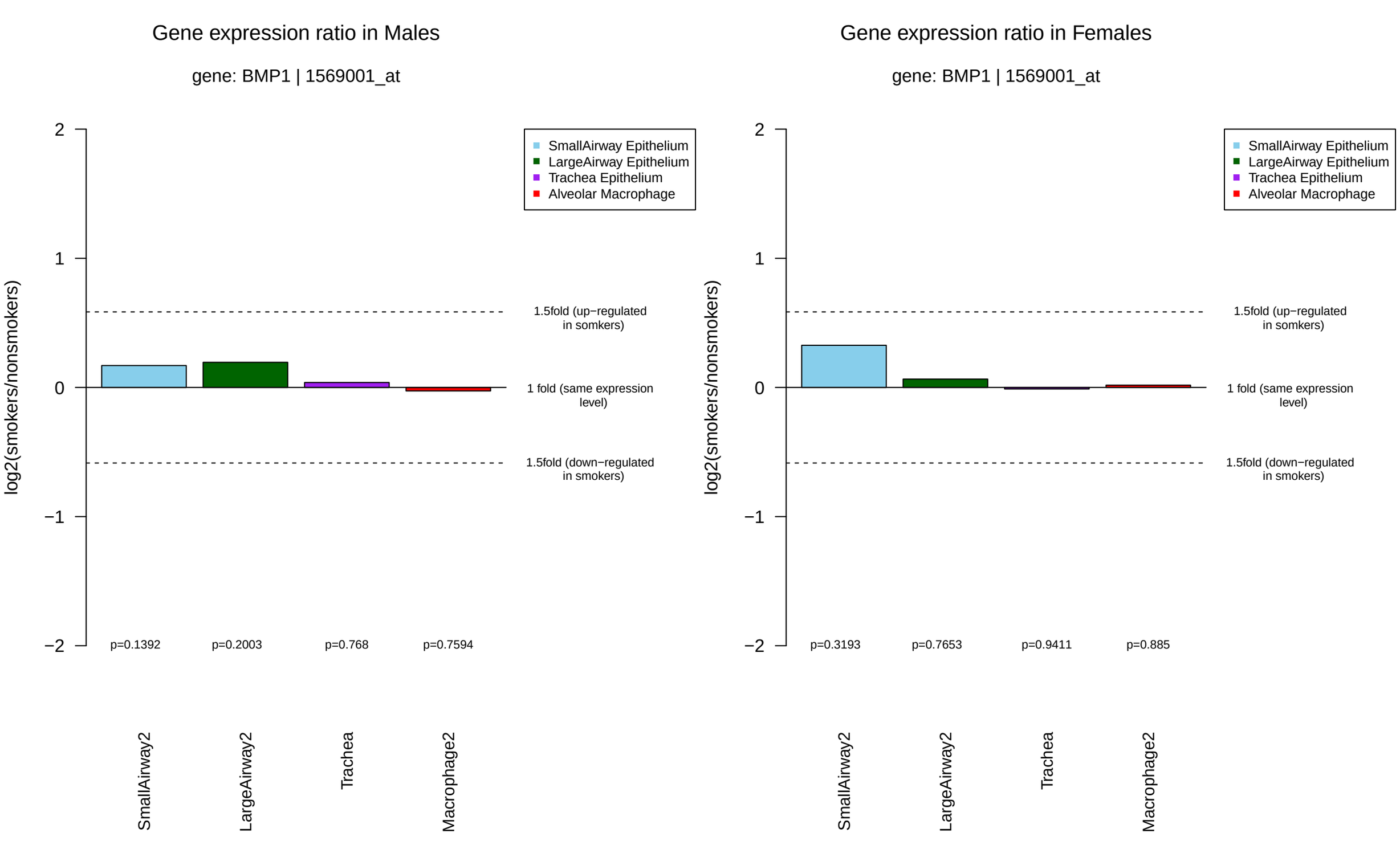
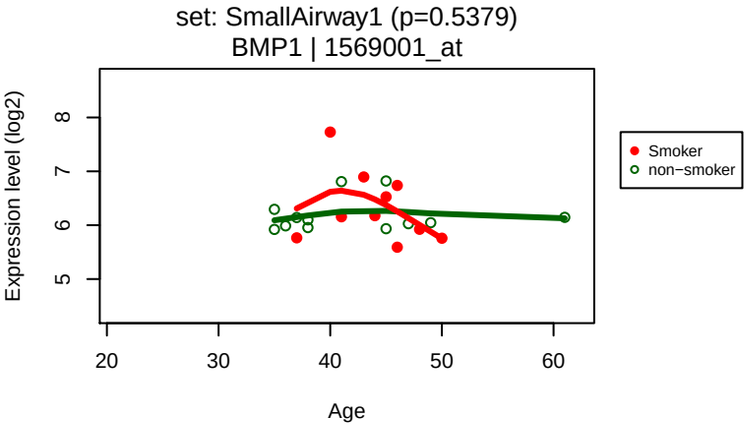
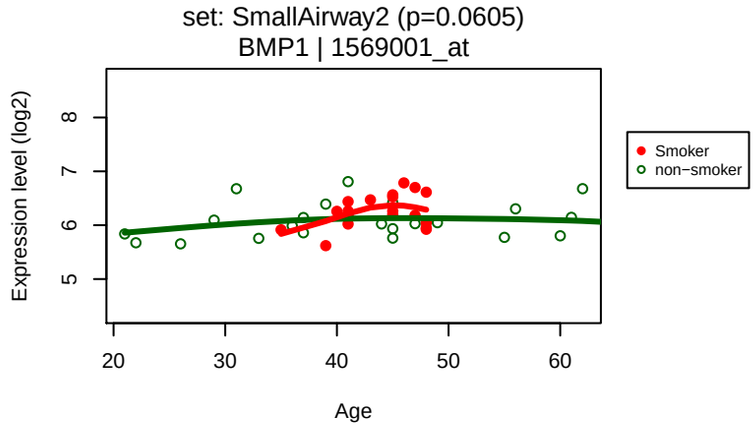
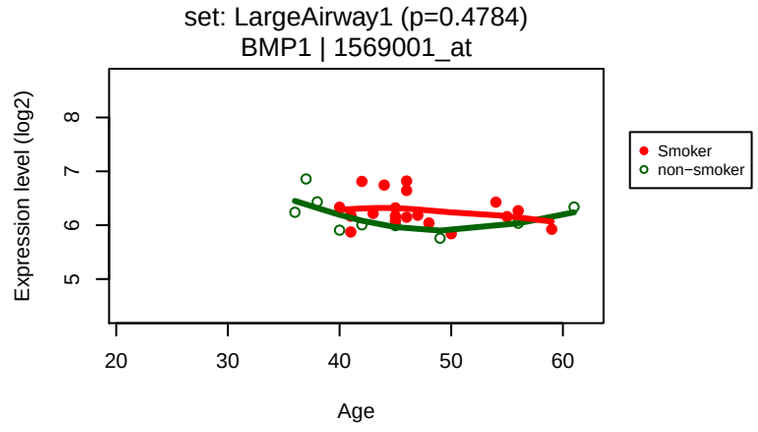

Gene expression of smokers and nonsmokers
gene: BMP1 | 1569001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5379
p=0.0605
p=0.0128
p=0.4784
p=0.1454
p=0.967
p=0.3969
p=0.8892
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

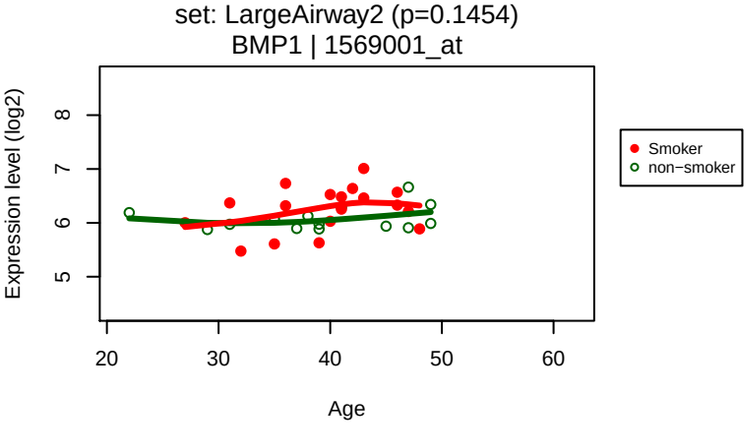

Gene expression ratio (smokers/nonsmokers)
gene: BMP1 | 1569001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5379
p=0.0605
p=0.0128
p=0.4784
p=0.1454
p=0.967
p=0.3969
p=0.8892
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

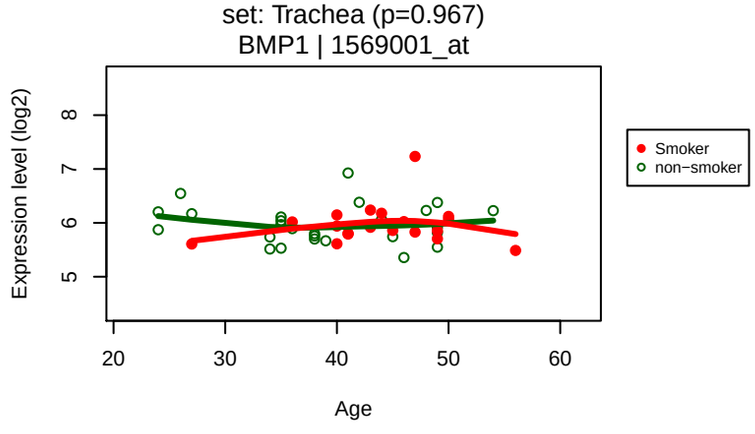

Gene expression ratio in Males
gene: BMP1 | 1569001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1392
p=0.2003
p=0.768
p=0.7594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP1 | 1569001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3193
p=0.7653
p=0.9411
p=0.885
SmallAirway2
LargeAirway2
Trachea
Macrophage2

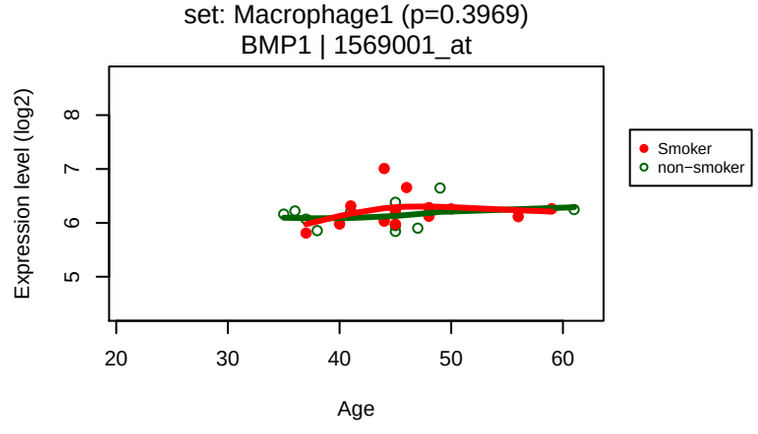

set: SmallAirway1 (p=0.5379)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker

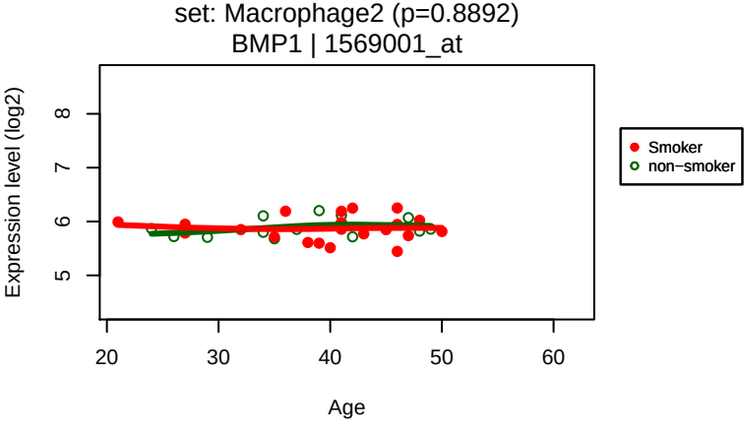

set: SmallAirway2 (p=0.0605)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker

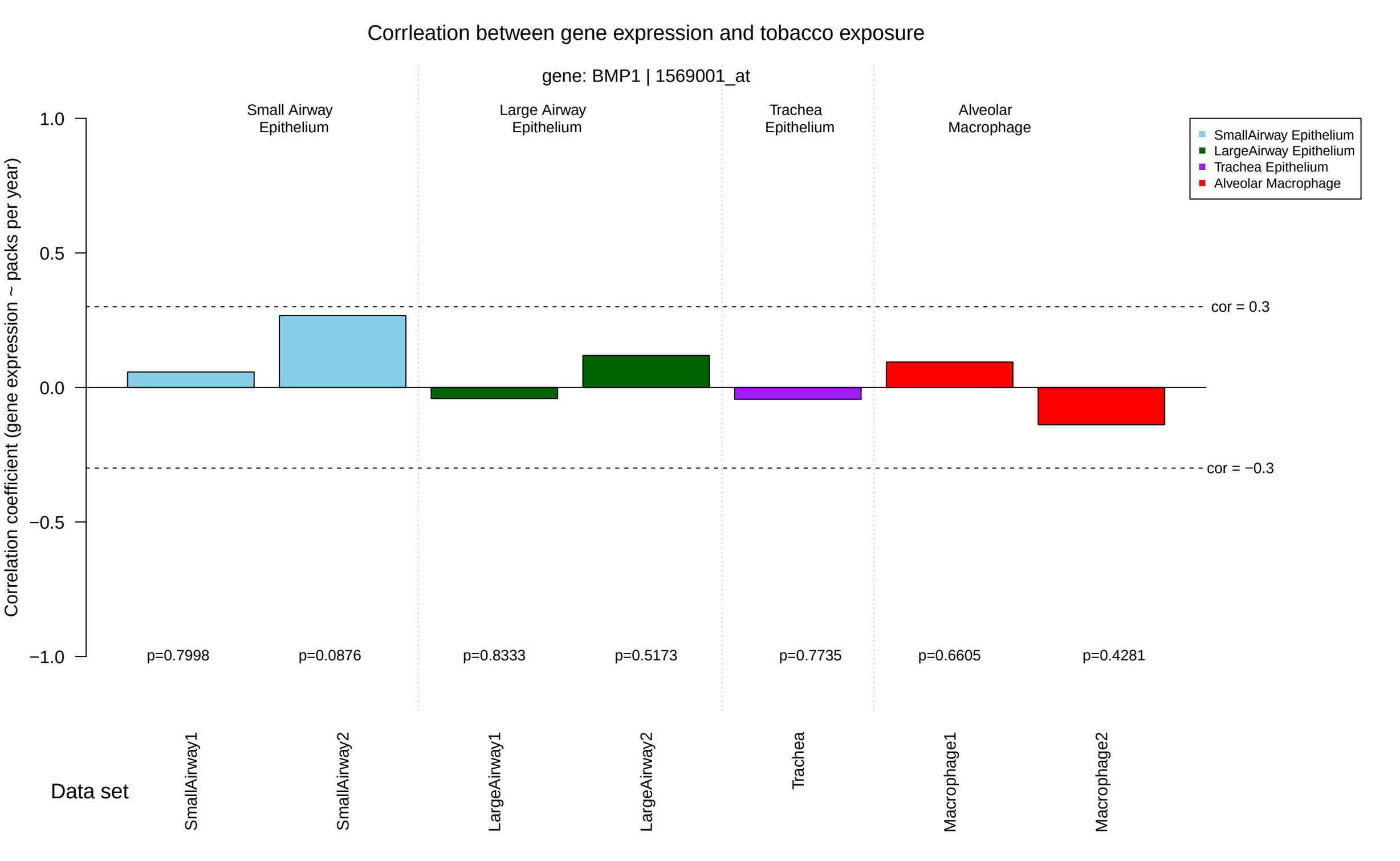

set: LargeAirway1 (p=0.4784)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1454)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.967)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3969)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8892)
BMP1 | 1569001_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP1 | 1569001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7998
p=0.0876
p=0.8333
p=0.5173
p=0.7735
p=0.6605
p=0.4281
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2